

Muhammad Ali Raza

COMPUTER VISION & FULL-STACK AI ENGINEER

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TECHNICAL SKILLS

LANGUAGES

Python, JavaScript, C++, SQL

COMPUTER VISION & ML

YOLOv11 / YOLO26 (Ultralytics), InsightFace, DeepSort, OpenCV, PyTorch (CUDA), TensorFlow / Keras, ONNX Runtime, scikit-learn, NumPy, pandas

ML TECHNIQUES

Transfer learning & model fine-tuning, dataset augmentation, vector embeddings, cosine-similarity search, anti-spoofing (PAD)

BACKEND

Django, Django REST Framework, PostgreSQL, Redis, Flask, Gunicorn, multithreading, REST API design

FRONTEND

React 19, Vite, Tailwind CSS, Chart.js / Recharts, Axios, React Router

TOOLING & DELIVERY

Git, PyInstaller, MediaMTX (RTSP), Selenium, Postman, Figma, Linux / Windows

EDUCATION

BS Computer Science

Riphah International University
Lahore, Pakistan · 2021 – 2026

COURSEWORK

Artificial Intelligence · Computer Vision · Data Structures & Algorithms · Database Systems · Web Engineering · Object-Oriented Programming · System Architecture

STRENGTHS

Real-time systems

Concurrency

GPU inference

End-to-end delivery

Problem solving

PROFILE

Computer Science graduate specializing in **real-time computer vision** and **full-stack engineering**. Designed and built **CodeWatch**, a multi-camera surveillance platform that detects, tracks, and identifies people across live video feeds and enforces access rules in real time. Comfortable across the whole stack — from GPU inference pipelines and model fine-tuning to REST APIs and production React dashboards — with a focus on shipping working, end-to-end systems.

FLAGSHIP PROJECT

CodeWatch — Real-Time Multi-Camera Surveillance & Access Control 2025 – 2026

Final Year Project · Architecture & full-stack development

Python, PyTorch, Django REST, React, Redis, PostgreSQL, OpenCV

- Built an end-to-end pipeline that ingests live **RTSP/MJPEG** feeds and, per frame, detects people (**YOLO26/YOLO11** segmentation), tracks them (**DeepSort**), and identifies them against a face database (**InsightFace** 512-d embeddings with vectorized cosine matching).
- Engineered a **multithreaded engine** (one thread per camera, shared model singletons guarded by locks) sustaining the full detect → track → identify → dress-code pipeline at ~5 analyzed FPS, scaling to **~8 cameras per GPU**.
- Fine-tuned a **custom YOLO dress-code detector** (15 clothing categories) via two-stage transfer learning with a frozen backbone, plus a data-augmentation pipeline to extract and re-label samples and close class gaps.
- Added a **MiniFASNet anti-spoofing** (presentation-attack-detection) module in TorchScript, benchmarked at **~3.6 ms/face**, to reject photo and replay spoofs before identity matching.
- Implemented **cross-camera identity hand-off** with Redis TTL keys so a person retains one identity while moving between cameras; added auto-blacklisting via DB signals and deduped violation logging.
- Designed the **Django REST** backend (PostgreSQL, Redis cache, role-based access, email/push notifications) and a **React 19 + Tailwind** operator dashboard with live feeds, analytics, movement heatmaps, and PDF reports.
- Packaged the full stack (CV engine, API, dashboard, RTSP server) into a **one-click system-tray desktop launcher** (PyInstaller / PyQt5) for non-technical operators.

EXPERIENCE

Generative AI Intern

2024

NETSOL Technologies · Lahore

- Built **Selenium** automation for large-scale web data scraping and extraction.
- Developed data-preprocessing pipelines — document **chunking and OCR** — to digitize source material for model training.
- Generated and managed **vector embeddings** to support training and retrieval for custom AI models.